



Annunciator Module



E10-0068

Installation and Operation Instructions

EPACO

Introduction

Fike is pleased to present a new Installation, Operation and Maintenance Manual for our Explosion Protection System. This document has been created to incorporate the most up-to-date information available for this Fike product and to make it easy to use.

Who should read this manual?

This manual is intended for those individuals who are custodians of the Fike EPACO™ System. Others such as architects, engineers, sales and marketing personnel, etc. will find the information useful as well.

Warranty Information

Fike provides a one-year limited manufacturer's warranty on the product identified in this manual. Copies of the warranty can be obtained from an authorized Fike sales outlet. An authorized Fike sales outlet, using the MRA procedure, must return warranty items. See Section 9 of this manual for details of returning product to Fike.

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The information contained in this manual is as accurate as possible. This manual is intended to be an aid to Fike authorized sales outlets or engineers charged with the installation and maintenance of the Fike Explosion Protection System. Fike does not warrant that this manual is technically correct, complete or the product referenced herein are free from minor flaws. Fike reserves the right to change the information contained in this manual without notice.

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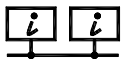
Table of Contents

1.0	Terms and Symbols Used In This Manual	1
2.0	EPACO System Overview	3
3.0	System Components	4
4.0	Installation	5
4.1	Installation Overview	5
4.2	Annunciator Module	6
4.3	Circuit Connections	7
4.3.1	Power Input (P3)	7
4.3.2	Trouble Relay (P4)	7
4.3.3	Remote Bus (P1)	8
5.0	Operation	10
5.1	Communication	10
5.1.1	Status Bus	10
5.1.2	Remote Bus	10
5.1.3	Computer Connection	10
5.2	AM Status	11
5.2.1	Normal State	11
5.2.2	Trouble State	11
5.2.3	Alarm State	11
5.3	LED Display	12
5.4	Operating Modes	12
5.4.1	Navigation	12
5.4.2	Main Menu	13
5.4.3	Status Operation	13
5.4.4	History Operation	20
5.4.5	Pressure Reading	22
5.4.6	Enable / Disable EPC	25
5.4.7	PSU Output On/Off	26
5.4.8	Enable / Disable RC8	27
5.4.9	Event History	28
6.0	Decommissioning Procedure / Check-List	29
7.0	Spare Parts	30
8.0	Specifications	31
9.0	Repair and Return Authorization	32
Annex A	EPACO Wiring Diagram	34
Annex B	Annunciator Module Menu	36

List of Exhibits

Exhibit 2-1 EPACO System Block Diagram.....	3
Exhibit 3-1 AM Photo	4
Exhibit 4-1 AM Block Diagram.....	6
Exhibit 4-2 AM Bottom View Wiring Connections.....	7
Exhibit 4-3 AM Power Input Wiring.....	7
Exhibit 4-4 Trouble / Supervisory Relay	7
Exhibit 4-5 EPC Top View – P1-P2 Wiring Connections.....	8
Exhibit 4-6 Remote Bus Wiring Diagram	8
Exhibit 4-7 Status Wiring Diagram.....	9
Exhibit 5-1 Status Bus	10
Exhibit 5-2 Remote Bus	10
Exhibit 5-3 Serial Port.....	10
Exhibit 5-4 Address Dip Switch	10
Exhibit 5-5 Power LED	11
Exhibit 5-6 Trouble LED	11
Exhibit 5-7 Alarm LED	11
Exhibit 5-8 Annunciator Module Pushbuttons	12
Exhibit 5-9 Annunciator Module Digits.....	12
Exhibit 5-10 Annunciator Module Main Menu Table.....	13
Exhibit 5-11 Status Menu Module Identifier Table.....	13
Exhibit 5-12 AM Status Codes.....	14
Exhibit 5-13 EPC Status Codes.....	14
Exhibit 5-14 PSU Status Codes.....	14
Exhibit 5-15 RC8 Status Codes.....	19
Exhibit 5-16 History Menu Top of File indicator.....	20
Exhibit 5-17 History Menu Bottom of File indicator	20
Exhibit 5-18 History Menu Navigation	20
Exhibit 5-19 Display History Codes	21
Exhibit 5-20 Annunciator Module Pressure Menu Tables	22
Exhibit 5-21 Pressure Correlation Chart.....	23
Exhibit 5-21 Pressure Correlation Chart (continued).....	24
Exhibit 5-22 Enable/Disable Menu Table	25
Exhibit 5-23 ENABLED Designation for Digits 2 and 3	25
Exhibit 5-24 DISABLED Designation for Digits 2 and 3	25
Exhibit 5-25 PSU Power Output On/Off Table	26
Exhibit 5-26 ON Designation for Digits 2 and 3.....	26
Exhibit 5-27 OFF Designation for Digits 2 and 3	26
Exhibit 5-28 RC8 Enable/Disable Menu Table	27
Exhibit 5-29 ENABLED Designation for Digits 2 and 3	27
Exhibit 5-30 DISABLED Designation for Digits 2 and 3	27
Exhibit 5-31 Event History Table	28
Exhibit 6-1 Decommissioning Steps	29
Exhibit 7-1 Spare Parts List	30
Annex A EPACO Wiring Diagram.....	35
Annex B AM Menu.....	37

1.0 TERMS AND SYMBOLS USED IN THIS MANUAL

<u>Term/Symbol</u>	<u>Description</u>
	Alarm symbol. This symbol is on the face of the EPC next to the red alarm LED. This LED will illuminate when the EPC has validated an alarm signal from a detection device, and as a result, initiated the actuators. ("Alarm" Red LED ON, Piezo sounding continuous) The alarm occurs when an input circuit configured for alarm operation has been activated. A detector or other contact device typically initiates activation.
	Trouble symbol. This symbol is on the face of the EPC next to the yellow trouble LED. This LED will illuminate on any trouble condition where the EPC's monitoring system is indicating a fault in one or more of its systems. The EPC may or may not be able to provide a response to an activation signal, depending upon the exact nature of the fault.
	DC Power symbol. This symbol is on the face of the EPC next to the green LED. This LED illuminates at a normal state to indicate that the unit is receiving power. This symbol is also located on the bottom of the AM. It designates the 24VDC input wiring location.
 NO NC C	Trouble Relay symbol. This symbol located on the bottom of the AM. It designates the relay output wiring connections.
	Remote Bus symbol. This circuit is a low speed network connection that sends status information from the Annunciator Modules to a computer for system monitoring. This network is provided to support future upgrades to the EPACO system – features associated with this network are currently not implemented.
	Status Bus symbol. This circuit is a low speed network connection that sends status information to other modules. This bus transmits "state" information to the zones.
	Fuse symbol. This designation represents a circuit protection fuse. The fuse is rated at the amperage noted next to this marking.
 DB9	Computer Connection symbol. This designation represents the location for the computer connection. Using a straight-through serial cable, the PSU can be connected from this DB9 connection to a computer serial port. The EP Works software can be installed on the computer and provide operational information.
Ω	Symbol for "ohm". Unit of resistance.
Protection Loop	All EPACO modules attached to the AM. A protection loop consists of a maximum of 4 EPCs, 1 PSU, 1 AM, and 4 RC8s.
Disabled State	The condition when the EPC is not allowed to initiate the actuators.

<u>Term/Symbol</u>	<u>Description</u>
Normal State	("Trouble" Yellow LED OFF) The system is in the normal state when the power supply and all circuits are properly configured, connected, and responding. The system remains in normal state until a trouble or alarm condition occurs.
Actuator	An actuator is the electrically actuated firing device used to initiate the closing of an isolation valve or discharge a suppression container. Typically actuators are a GCA (Gas Cartridge Actuator) or detonators.
Cross-zone Detection	Detection scheme where two detectors must activate before the system releases. The first detector does not latch.

2.0 EPACO SYSTEM OVERVIEW

The EPACO™ System, is a modular detection and control system combining the latest in addressable technology with simplicity of installation and maintenance. All system modules are DIN rail mounted to allow for a variety of installation options. Three (3) bus type communication circuits tie the various system modules into one easy to operate protection system. A non-volatile history buffer allows for enhanced diagnostic ability to troubleshoot process situations. With the optional

Annunciator Module (AM), the customer has the ability to step through a menu format to retrieve process history without having to wait for a service agency to arrive on site.

This manual will provide the necessary information to properly install and monitor the Annunciator Module (AM) for the EPACO™ System. The commissioning and service should be performed by certified service engineers and outlets specifically trained on this system.

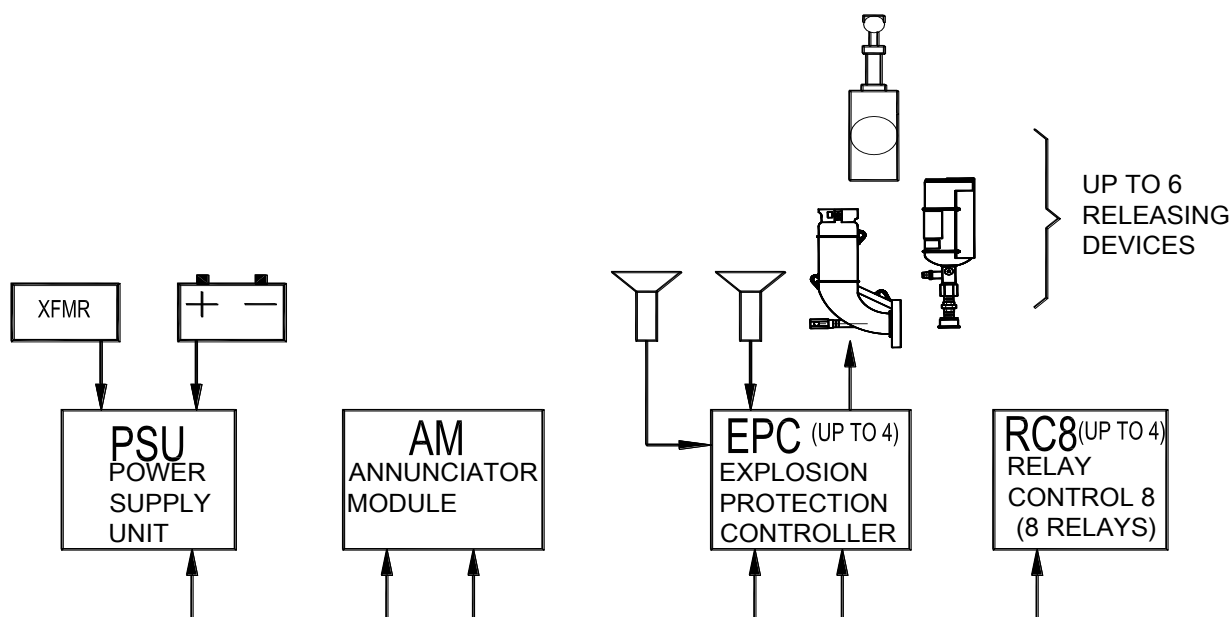


Exhibit 2-1 EPACO System Block Diagram

3.0 SYSTEM COMPONENTS

The cornerstone of the system is the Explosion Protection Controller (**EPC**). Three colored LEDs provide instant visual indication of system status. The DIN rail mounting allows for flexibility when choosing an enclosure. The EPC is an addressable panel that has the ability to retain an event history for enhanced system diagnostics. The EPC is a component that allows mounting in close proximity to the protected environment, thereby minimizing field wiring. Shorter wire runs greatly reduce the interference from electrical and radio frequency sources, allowing for a much more reliable protection system. The EPC's detection inputs can be programmed for pressure warning, threshold detection, and rate of rise detection. It also has a contact closure detection circuit to support thermal, infrared, or other switch-closure type detection devices¹. A supervisory input circuit is provided to monitor suppression container pressure or other similar system status safeguards. The EPC has a remote disable input contact to allow for disabling the EPC from a customer PLC or other remote device during product loading, cleaning, or maintenance. DIP switches allow for basic system programming. An RS232 connection is available at each module for connection to a PC. Using the EP Works™ software, an authorized user can access system diagnostics and more complex programming by PC.

The EPC is powered by the other key component in the EPACO™ system, the Power Supply Unit (**PSU**). The PSU has six separate 24 VDC, 1 amp power output circuits capable of powering up to four EPCs in addition to other components of the EPACO™ system. The PSU has the flexibility of containing its own battery backup supply or the customer has the option of supplying a backup power source. The PSU has an imbedded power shutdown procedure in the event of AC power failure, which prevents the backup batteries from being completely depleted. In addition to powering EPCs, the PSU can power an Annunciator Module (AM) and Relay Card Modules (RC8). The power circuit dedicated for the Annunciator Module can not be shut off. All other power output circuits can be shut off for service on the various system components. The PSU may be eliminated if

battery backed, uninterrupted 24VDC, 0.500 Amps power per module can be provided by others. For an FM approved installation without the PSU, the uninterruptible power must be U.L. 1778 "Uninterruptible Power Systems" listed, and conform to NFPA 72 "National Fire Alarm Code", NFPA 70 "National Electrical Code" (Articles 700 and 701), NFPA 110 "Standard for Emergency and Standby Power Systems", and NFPA 111 "Standard on Stored Electrical Energy Emergency and Standby Power Systems".

The Annunciator Module (**AM**) provides a central communication point for diagnostics. This module will typically be installed in an area removed from the process environment such as a control room. The AM has two push buttons to access the various menus and three numbered LEDs for identification of the system statuses within the menus.

The Relay Card Module (**RC8**) provides the user with a block of eight form "C" relays for equipment shutdowns and remote notification of system trouble and alarm conditions. On-board LEDs display status for instant visual verification of each relay. DIP switches are provided for configuring the relay's operation.



Exhibit 3-1 AM Photo

¹ FM Approved switch closure for use with Fike threshold detector (E61-042-x) and Rate of Rise detector (E61-056-x). Other switch closure, thermal, or infrared devices are not approved by FM.

4.0 INSTALLATION

4.1 INSTALLATION OVERVIEW

The effectiveness of active explosion protection systems, such as explosion suppression and explosion isolation, depends upon the instantaneous reaction of the protection system and is a direct function of its speed of response.

It is therefore, critical that all possible measures are taken to reduce the individual system components' response times to an absolute minimum.

An active explosion protection system basically consists of three components: One or more explosion detectors, an electronic system controller, and one or more protective devices such as explosion suppressors or isolation valves. Instrumentation wiring interconnects these components.

The system controller is microprocessor-based and shall be installed in a location that maintains the EPC's temperature rating of -18°C to 43°C (0°F to 110°F) when it is installed in an auxiliary housing. For FM Approved installations the housing shall be a lockable enclosure conforming to the installed area requirements as defined by NEMA 250 "Enclosures for Electrical Equipment (1000 Volts Maximum)."

Besides its function as a fire controller it also incorporates an event table and a self-checking feature to continuously monitor the complete system-loop for errors or system defects.

Electronic devices, microprocessors, pyrotechnic initiators, and field wiring are influenced by the electromagnetic "environment" surrounding these components. The use of cellular telephones, transmitters, induction motors, welding equipment or the presence of power cables and transformers can create environments with high levels of electromagnetic radiation, resulting in induced electrical "noise" or voltage peaks.

Such effects are known to designers and manufacturers of instrumentation and control systems (PLC's), used in industrial environments and are handled through the use of specially designed electronic filters. These filters neutralize the unwanted noise and offer a "clean" signal for further processing. The filters, however, result in a delay in the processing of signals, and can therefore only be applied with great care in explosion protection systems where the effectiveness depends on the overall response time.

In active explosion protection systems, a balance must be maintained: The system must be extremely fast to achieve the required effectiveness, but at the same time must be stable and insensitive to surrounding sources of noise.

The system controller will detect and report major system troubles (such as ground faults, wire disconnection, and unstable input or output signals) and indicates the need for appropriate action.

The system controller will also detect unacceptable levels of electromagnetically induced noise. If the magnitude of the noise is such that this may result in a risk for spontaneous system activation or affect the system's performance, the controller will revert to its default error-mode.

It is essential to practice extreme caution when selecting component location, cable specifications, cable routes, and the "cleanliness" of the offered power source. In order to reduce the electromagnetic induced noise to a level that will not affect the required performance of the explosion protection system, verify all earth connections. It is preferred to have the enclosure and conduits connected to Protective Earth (similar with other building grounds) while the drain wires from the field wiring and each module ground connected to a separate Instrument Earth. This Instrument Earth connection shall not have inductive or capacitive loading such as motors, welders, or other industrial equipment. Where a separate earth connection is not available, the drain wires and module ground connections should be made to battery common. Complying with the following recommendations will help minimize the induced noise to acceptable levels.

4.2 ANNUNCIATOR MODULE

The Annunciator Module (AM) is the primary system status indication point for the EPACO control system. It provides a single point for retrieving history as well as system status for all EPACO modules attached to the Remote Bus.

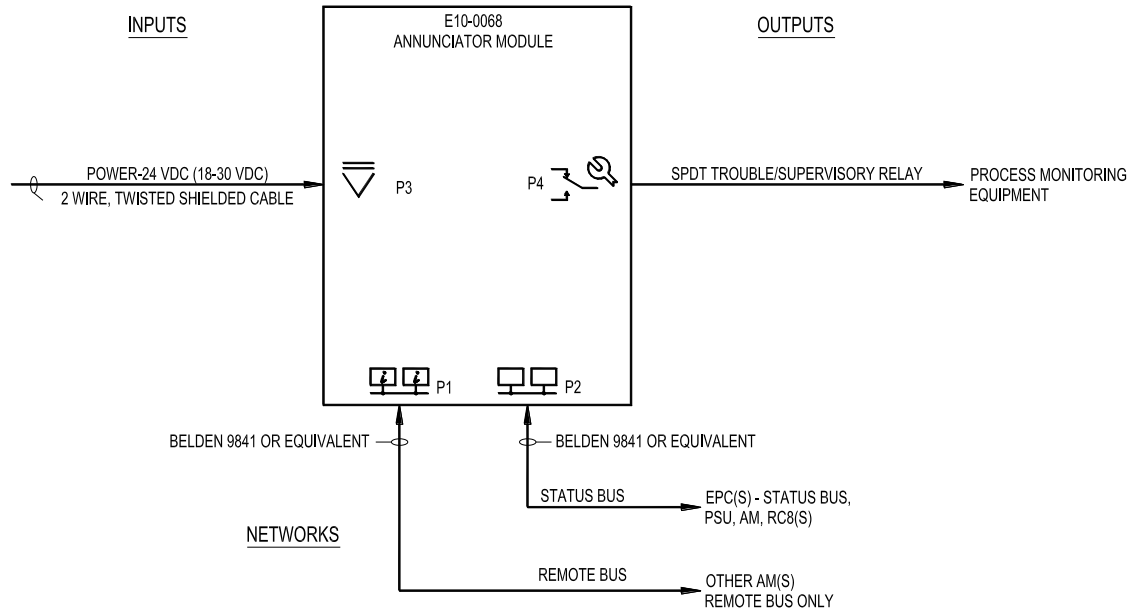


Exhibit 4-1 AM Block Diagram

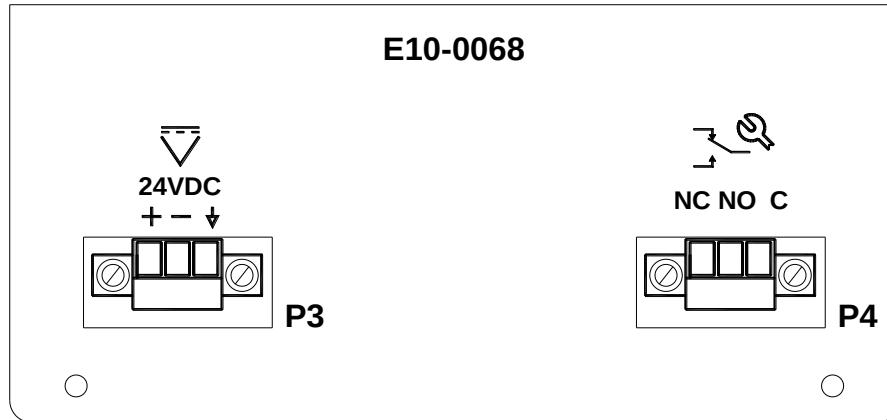


Exhibit 4-2 AM Bottom View Wiring Connections

4.3 CIRCUIT CONNECTIONS

4.3.1 Power Input (P3)

Power Input requires between 18 to 30 Vdc, at 500mA. For the Annunciator Module, power needs to be uninterruptible for proper operation of the status bus.

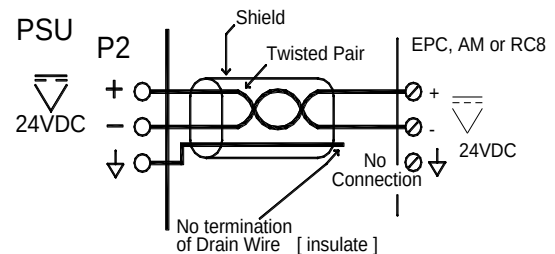


Exhibit 4-3 AM Power Input Wiring

4.3.2 Trouble Relay (P4)

The trouble relay is available to provide AM status to monitoring equipment. The trouble relay is a “fail-safe” relay that will transfer upon any system trouble or supervisory condition. This can be used for equipment shutdown, visual, audible, or other type annunciation. Trouble conditions may or may not affect the protection system’s ability to respond to a deflagration, but must be acknowledged and identified prior to continuing to run the process.

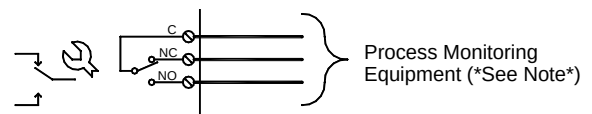


Exhibit 4-4 Trouble / Supervisory Relay



Note: The relay designations (NO, NC, and C) are shown in the de-energized state with no power applied. When the EPC is powered and the controller is in the NORMAL state, the trouble relay is energized. (The trouble relay marking is opposite of what is expected).

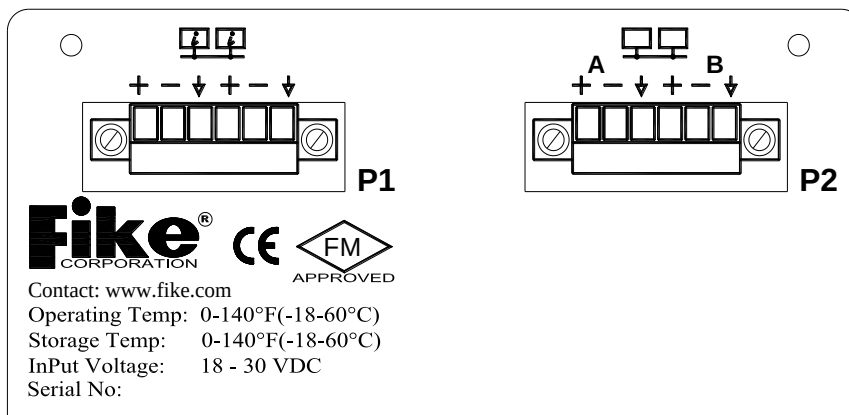


Exhibit 4-5 EPC Top View – P1-P2 Wiring Connections

4.3.3 Remote Bus (P1)

The hardware for the remote bus is provided for future planned upgrades to the EPACO system. Once these upgrades are implemented, a PC will

be capable of monitoring each of the protection loops via the attached Annunciator Modules. Refer to Exhibit 4-6 for the wiring diagram of the remote bus.

EP Control System - Remote Bus (future state)

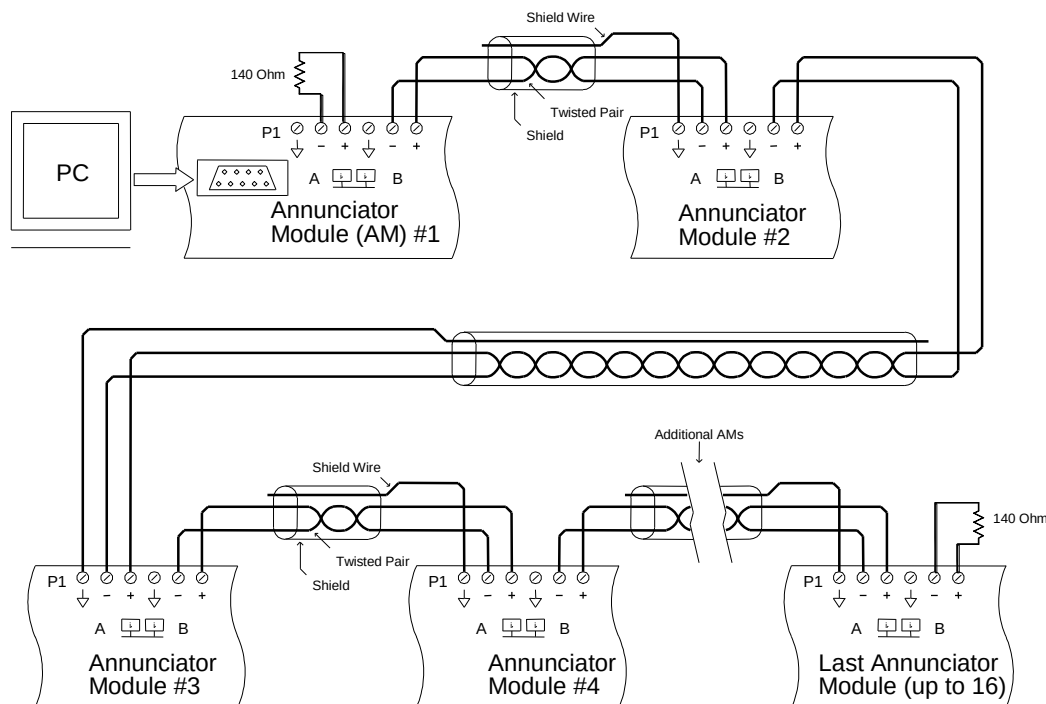


Exhibit 4-6 Remote Bus Wiring Diagram

4.3.4 Status Bus Connection (P2)

The Status Bus is a low speed communication bus that transmits control information between the EPC, Power Supply, Annunciator Module and Relay Modules. Belden 9841 or RE-Y2Y cabling is recommended for this circuit. Maximum resistance $R=50\ \Omega$, inductance $L=100\mu\text{H}$, and capacitance $C=0.02\mu\text{F}$ with a maximum length of 300m (1,000 ft).



Note: The B+ connects to the A+ on the next device; similarly the B- connects to the A- of the next device. Install the $140\ \Omega$, 1/2 watt resistor at each end as shown.



Note: The drain wire should only be connected on one end of each wire run. Make sure that the drain wire, which is not connected to a terminal, is cut and insulated from making contact to metal or other wiring connections.

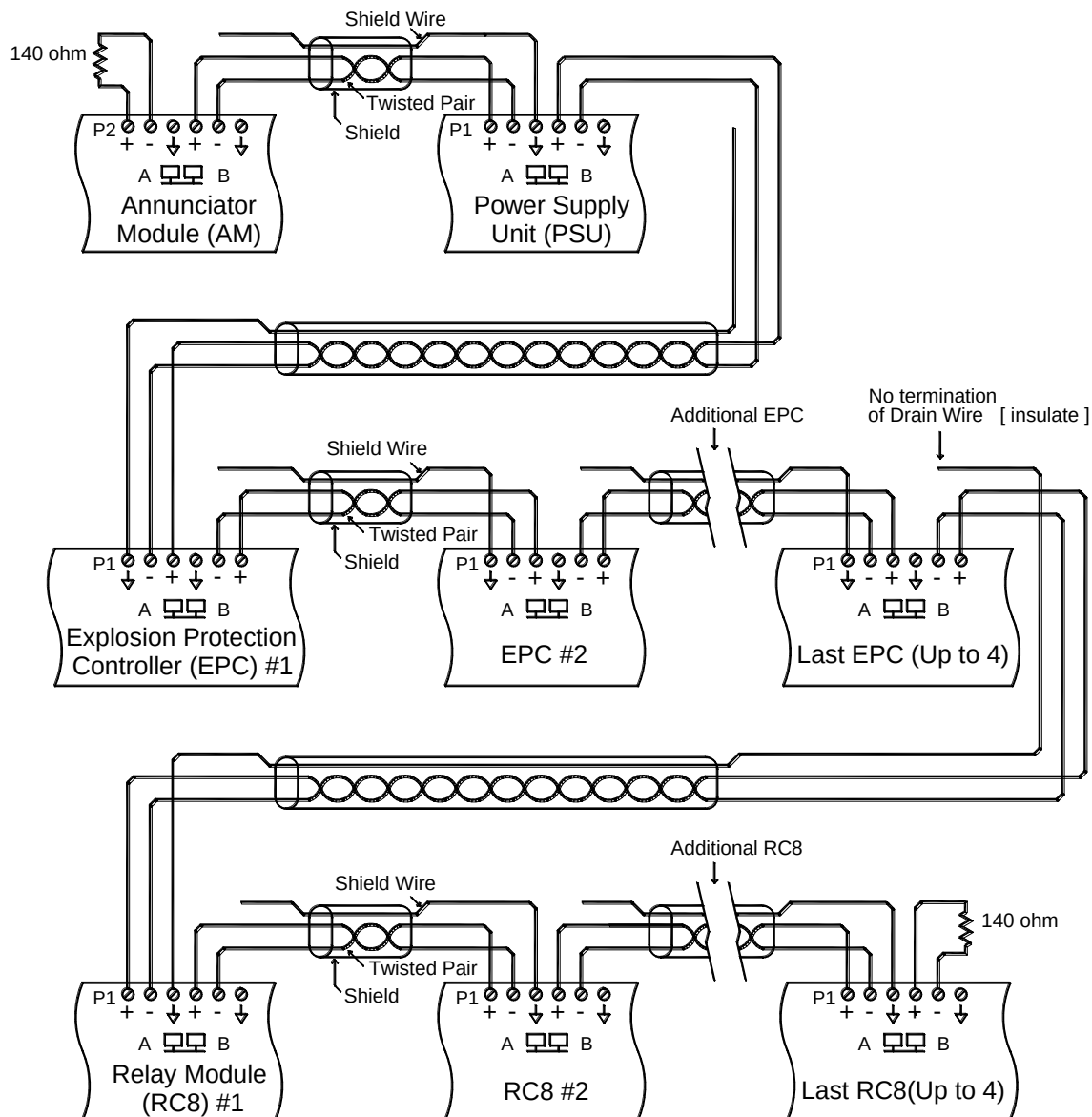


Exhibit 4-7 Status Wiring Diagram

5.0 OPERATION

5.1 COMMUNICATION

5.1.1 Status Bus

The EPACO System has three separate communication circuits, or buses. The Annunciator Module uses only the Status Bus and Remote Bus. The status bus ties all the various modules together to form a network for reporting of system status to the Annunciator Module.



Exhibit 5-1 Status Bus

5.1.2 Remote Bus

The remote bus interconnects all of the Annunciator Modules for central annunciation of all protected loops at a permanently attached computer. When utilizing multiple Annunciator Modules with EP protection loops each of the Annunciator Modules require addressing via the 4 ADD dipswitches depicted below. The Address switch is not applicable when there is only a single AM in the EPACO system Remote Bus.

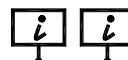


Exhibit 5-2 Remote Bus

5.1.3 Computer Connection

The EPC, the AM, and the RC8 are all programmable modules. System configuration can be performed on these modules by way of a DB9 serial port connection to a PC using EPWorks™ Software. Two levels of software are utilized for the EPACO™ system. Level one is a read-only user interface for the customer to check the status/history of his system. Level two is used by certified service technicians to program and service customer installations.

The ADD switch located on the front of the module provides the ability to address the AM when using multiple EPACO remote bus connections. Each AM on the Remote Bus shall be identified with a different address.



NOTE: Address and configuration changes are to be made by Fike authorized engineers or technicians only.

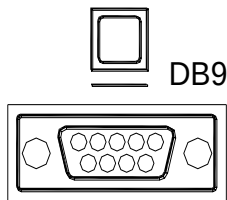


Exhibit 5-3 Serial Port

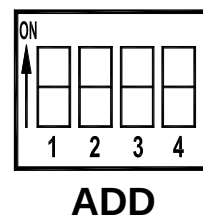
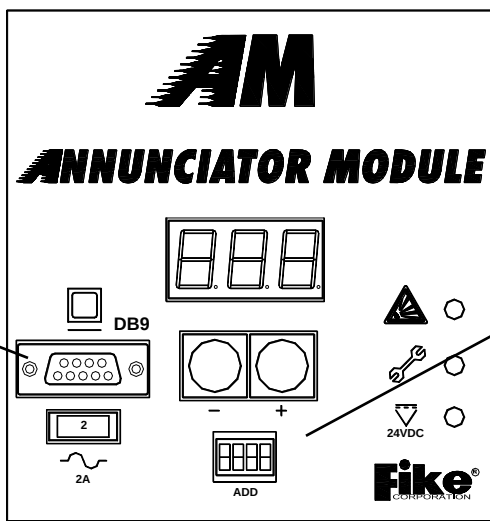


Exhibit 5-4 Address Dip Switch

5.2 AM STATUS

The AM is the customer interface for the EPACO system. The following tasks are accomplished by the AM:

- Monitors all EPC's, RC8's and PSU connected to the Status Bus and displays general status
- Provides a central location for history response
- Provides central location to connect computer and monitor history for entire EPACO system
- Communicates system status via trouble relay for user interface
- Real time pressure reading of the EPC's CEREx pressure transducers

The EPACO operation can be classified into three main states of operation: Normal, Trouble and Alarm. Each of these states is described in sections 5.2.1 through 5.2.3.

5.2.1 Normal State

When power is applied to the system, the EPACO system performs a 4 second initialization. At the end of the initialization, the trouble LED blinks slowly and the local piezo beeps in long recurring sound. It can then enter the "NORMAL" state if the power is of an appropriate voltage and there are no troubles or alarms. In the Normal State the green Power LED is 'ON', the local piezo is 'OFF', the trouble relay is energized. All other LED's are 'OFF'.

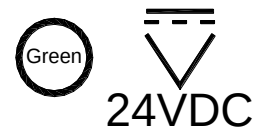


Exhibit 5-5 Power LED

5.2.2 Trouble State

A trouble occurs on the EPACO system if any one of the supervised devices experiences a wiring fault open or short condition, if the EPC configuration is invalid, if the system is disabled, if a process pressure warning level is exceeded, or if the input voltage drops below 18VDC. In the Trouble State the yellow Trouble LED is 'ON', the local piezo is 'ON', and the trouble relay is de-energized. Depending on the cause of the trouble, the system may or may NOT be completely functional. Each trouble should be investigated to determine the cause and promptly fixed. Each trouble, except for remote disable, will latch at the EPC. If the trouble is resolved, the EPC requires cycling of the power input to clear the trouble. If the remote disable is active, the EPC will enter the trouble state. When the remote disable is returned to normal, the EPC will automatically clear to the normal state without cycling the power.



Exhibit 5-6 Trouble LED



Note: If power is cycled to an EPC with a Fire Bus connected all EPCs connected to the Fire Bus will also need to have their power cycled simultaneously to restore the Fire Bus.

5.2.3 Alarm State

When the detection circuits have exceeded the alarm conditions required by the configuration, the EPC enters the Alarm State. In the Alarm State the red Alarm LED is 'ON', the local piezo 'ON', the P6 Alarm relay is de-energized, and the series firing output is activated. The alarm state is latching. If the alarm is cleared, the EPC requires cycling of the power input to clear the trouble.



Note: The alarm state will activate the system. If the system is reset after an alarm, it is typical to have an open wiring fault on the Series Firing circuit, since the GCA has activated.

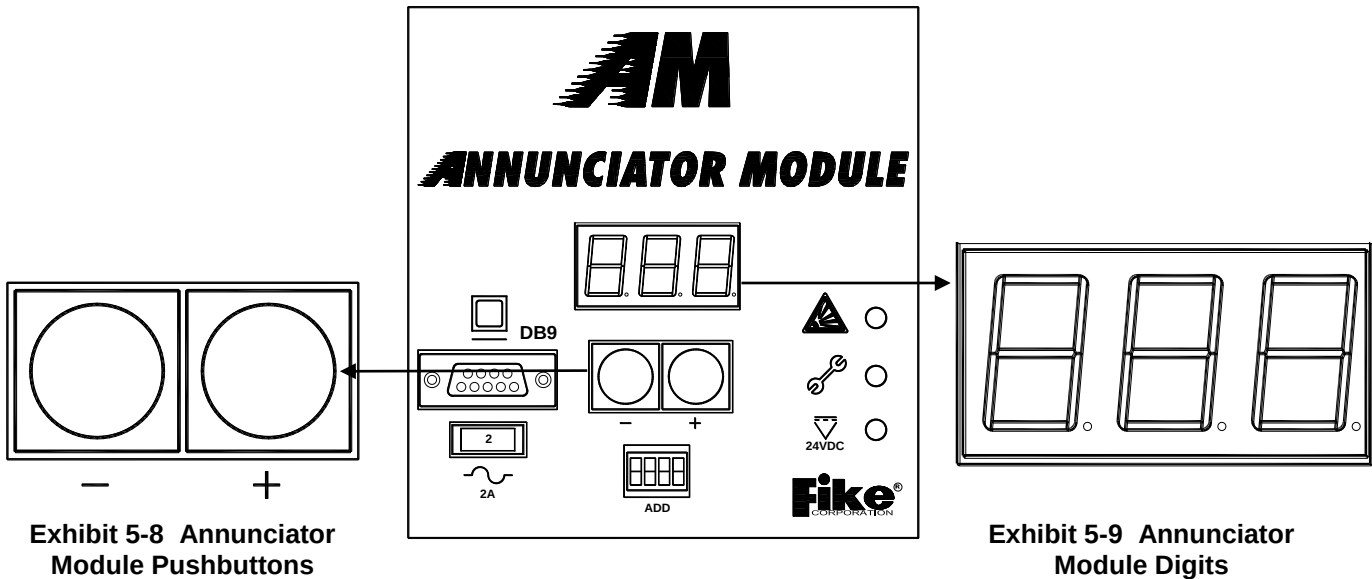


Exhibit 5-7 Alarm LED

5.3 LED DISPLAY

The Annunciator Module is a menu driven device, easily controlled by the two pushbutton switches on the front of the module.

All information is depicted on the 3 character LED display. All notations to follow will contain specific references to Digit 1, Digit 2, and Digit 3 of this display.



5.4 OPERATING MODES

There are six operating modes that the Annunciator Module utilizes:

- Status
- History
- Pressure Reading
- EPC Enable/Disable
- PSU Output On/Off
- Relay Card (RC8) Enable/Disable

During normal operating conditions, the Annunciator Module will default into the *Status* mode.

5.4.1 Navigation

Navigation through the Annunciator menus is easily accomplished while using the following rules:

1. Momentarily pressing both switches simultaneously will display the current operating mode of the Annunciator Module.
2. Pressing and holding both switches simultaneously for at least 5 seconds will allow the user access to the various operating modes (Menu Mode) of the Annunciator Module.
3. Once the Menu Mode of operation is entered, the left key is used to scroll between the menu options, and the right key is used to select the desired option.
4. The Menu Mode will timeout after 20 seconds if no activity is present.

5.4.2 Main Menu

Upon entry into the Main Menu, the following operational modes may be accessed:

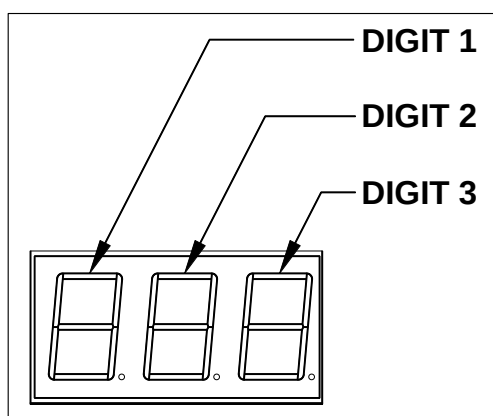


Exhibit 5-10 Annunciator Module Main Menu Table

Digit Number

1	2	3	Option Selected
S	t	A	STATUS
H	I	S	HISTORY
d	E	t	PRESSURE READING
E	P	C	EPC ENABLE / DISABLE
P	S	U	ON / OFF
r	C	B	RC8 ENABLE/DISABLE

In order to scroll from STA to any other option, press the left pushbutton until the desired operational mode is displayed, then select that mode by pressing the right pushbutton.

5.4.3 Status Operation

The Status Menu is provided to annunciate the current (most recent) operating condition of each of the EPACO system modules. In order to view the current status of each system module, scroll from device to device using the left pushbutton to decrement from 9 to 0 and the right pushbutton to increment from 0 to 9.

Digit 1 is used to identify the system module. Digits 2 and 3 are used to indicate the status of the selected module per their specific status table (described in the next pages).

Module Identifier

Digit Number Designates Module	Designates Status of Selected Module		Module Selected
1	2	3	
0	X	X	Annunciator Module
1	X	X	EPC #1
2	X	X	EPC #2
3	X	X	EPC #3
4	X	X	EPC #4
5	X	X	RC8 #1
6	X	X	RC8 #2
7	X	X	RC8 #3
8	X	X	RC8 #4
9	X	X	Power Supply Unit

Exhibit 5-11 Status Menu Module Identifier Table

AM Status Codes *Note: The Annunciator will display only the highest priority code active.*

Digit Number		Designates Status of Selected Module		Annunciator Module Status
Designates Module		2	3	
0 = AM				
0	0	1		Normal Status Bus Loop
0	0	2		Module Trouble on Status Bus
0	0	3		EP System on Status Bus Loop Activated
0	0	4		Communication Failure on Status Bus Loop
0	0	5		PC Mode

Exhibit 5-12 AM Status Codes**EPC STATUS CODES**

Digit Number		Designates Status of Selected Module		EPC Status
Designates Module		2	3	
1-4 = EPC				
X	-	-		No EPC or EPC Shutdown
X	0	1		Normal Operation Mode ←lowest priority
X	0	2		Trouble on one of the Input Circuits
X	0	3		Pressure Warning Level Exceeded
X	0	4		Trouble on one of the Output Circuits
X	0	5		Trouble on the Fire Bus
X	0	6		PredischARGE Condition Satisfied
X	0	7		Supervisory Input Active
X	0	8		EPC has Activated Releasing Outputs
X	0	9		EPC has been Disabled
X	1	0		Failure during Arming ←highest priority

Exhibit 5-13 EPC Status Codes**PSU STATUS CODES**

Digit Number		Designates Status of Selected Module		PSU Status
Designates Module		2	3	
9 = PSU				
9	-	-		No PSU/External Power Supply used or No Communications
9	0	1		Normal Operation Mode ←lowest priority
9	0	2		Fan Trouble
9	0	3		Ground Fault Trouble
9	0	4		Battery or Charge Trouble
9	0	5		AC Trouble
9	0	6		Battery Backup being Utilized
9	0	7		Output Trouble
9	0	8		Outputs Disabled
9	0	9		Auto Shutdown Mode ←highest priority

Exhibit 5-14 PSU Status Codes

RC8 MODULE STATUS CODES

Digit Number			STATUS OF RELAY							
Designates Module	Designates Status of Selected Module		WITH CODE AT LEFT DISPLAYED ON ANNUNCIATOR MODULE							
5-8 = RC8	2	3	1	2	3	4	5	6	7	8
5	WRITE IN									
6	STATE									
7	PROGRAMMED									
8	TO RIGHT									
X	-	-	No RC8 or No Communications							
X	0	0	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
X	0	1	ON	OFF	OFF	OFF	OFF	OFF	OFF	OFF
X	0	2	OFF	ON	OFF	OFF	OFF	OFF	OFF	OFF
X	0	3	ON	ON	OFF	OFF	OFF	OFF	OFF	OFF
X	0	4	OFF	OFF	ON	OFF	OFF	OFF	OFF	OFF
X	0	5	ON	OFF	ON	OFF	OFF	OFF	OFF	OFF
X	0	6	OFF	ON	ON	OFF	OFF	OFF	OFF	OFF
X	0	7	ON	ON	ON	OFF	OFF	OFF	OFF	OFF
X	0	8	OFF	OFF	OFF	ON	OFF	OFF	OFF	OFF
X	0	9	ON	OFF	OFF	ON	OFF	OFF	OFF	OFF
X	0	A	OFF	ON	OFF	ON	OFF	OFF	OFF	OFF
X	0	B	ON	ON	OFF	ON	OFF	OFF	OFF	OFF
X	0	C	OFF	OFF	ON	ON	OFF	OFF	OFF	OFF
X	0	D	ON	OFF	ON	ON	OFF	OFF	OFF	OFF
X	0	E	OFF	ON	ON	ON	OFF	OFF	OFF	OFF
X	0	F	ON	ON	ON	ON	OFF	OFF	OFF	OFF
X	1	0	OFF	OFF	OFF	OFF	ON	OFF	OFF	OFF
X	1	1	ON	OFF	OFF	OFF	ON	OFF	OFF	OFF
X	1	2	OFF	ON	OFF	OFF	ON	OFF	OFF	OFF
X	1	3	ON	ON	OFF	OFF	ON	OFF	OFF	OFF
X	1	4	OFF	OFF	ON	OFF	ON	OFF	OFF	OFF
X	1	5	ON	OFF	ON	OFF	ON	OFF	OFF	OFF
X	1	6	OFF	ON	ON	OFF	ON	OFF	OFF	OFF
X	1	7	ON	ON	ON	OFF	ON	OFF	OFF	OFF
X	1	8	OFF	OFF	OFF	ON	ON	OFF	OFF	OFF
X	1	9	ON	OFF	OFF	ON	ON	OFF	OFF	OFF
X	1	A	OFF	ON	OFF	ON	ON	OFF	OFF	OFF
X	1	B	ON	ON	OFF	ON	ON	OFF	OFF	OFF
X	1	C	OFF	OFF	ON	ON	ON	OFF	OFF	OFF
X	1	D	ON	OFF	ON	ON	ON	OFF	OFF	OFF
X	1	E	OFF	ON	ON	ON	ON	OFF	OFF	OFF
X	1	F	ON	ON	ON	ON	ON	OFF	OFF	OFF
X	2	0	OFF	OFF	OFF	OFF	OFF	ON	OFF	OFF
X	2	1	ON	OFF	OFF	OFF	OFF	ON	OFF	OFF
X	2	2	OFF	ON	OFF	OFF	OFF	ON	OFF	OFF
X	2	3	ON	ON	OFF	OFF	OFF	ON	OFF	OFF
X	2	4	OFF	OFF	ON	OFF	OFF	ON	OFF	OFF
X	2	5	ON	OFF	ON	OFF	OFF	ON	OFF	OFF
X	2	6	OFF	ON	ON	OFF	OFF	ON	OFF	OFF
X	2	7	ON	ON	ON	OFF	OFF	ON	OFF	OFF
X	2	8	OFF	OFF	OFF	ON	OFF	ON	OFF	OFF
X	2	9	ON	OFF	OFF	ON	OFF	ON	OFF	OFF
X	2	A	OFF	ON	OFF	ON	OFF	ON	OFF	OFF
X	2	B	ON	ON	OFF	ON	OFF	ON	OFF	OFF
X	2	C	OFF	OFF	ON	ON	OFF	ON	OFF	OFF
X	2	D	ON	OFF	ON	ON	OFF	ON	OFF	OFF
X	2	E	OFF	ON	ON	ON	OFF	ON	OFF	OFF
X	2	F	ON	ON	ON	ON	OFF	ON	OFF	OFF
X	3	0	OFF	OFF	OFF	OFF	ON	ON	OFF	OFF
X	3	1	ON	OFF	OFF	OFF	ON	ON	OFF	OFF
X	3	2	OFF	ON	OFF	OFF	ON	ON	OFF	OFF
X	3	3	ON	ON	OFF	OFF	ON	ON	OFF	OFF
X	3	4	OFF	OFF	ON	OFF	ON	ON	OFF	OFF
X	3	5	ON	OFF	ON	OFF	ON	ON	OFF	OFF
X	3	6	OFF	ON	ON	OFF	ON	ON	OFF	OFF
X	3	7	ON	ON	ON	OFF	ON	ON	OFF	OFF
X	3	8	OFF	OFF	OFF	ON	ON	ON	OFF	OFF

Digit Number			STATUS OF RELAY							
Designates Module	Designates Status of Selected Module		WITH CODE AT LEFT DISPLAYED ON ANNUNCIATOR MODULE							
5-8 = RC8	2	3	1	2	3	4	5	6	7	8
X	3	9	ON	OFF	OFF	ON	ON	ON	OFF	OFF
X	3	A	OFF	ON	OFF	ON	ON	ON	OFF	OFF
X	3	B	ON	ON	OFF	ON	ON	ON	OFF	OFF
X	3	C	OFF	OFF	ON	ON	ON	ON	OFF	OFF
X	3	D	ON	OFF	ON	ON	ON	ON	OFF	OFF
X	3	E	OFF	ON	ON	ON	ON	ON	OFF	OFF
X	3	F	ON	ON	ON	ON	ON	ON	OFF	OFF
X	4	0	OFF	OFF	OFF	OFF	OFF	OFF	ON	OFF
X	4	1	ON	OFF	OFF	OFF	OFF	OFF	ON	OFF
X	4	2	OFF	ON	OFF	OFF	OFF	OFF	ON	OFF
X	4	3	ON	ON	OFF	OFF	OFF	OFF	ON	OFF
X	4	4	OFF	OFF	ON	OFF	OFF	OFF	ON	OFF
X	4	5	ON	OFF	ON	OFF	OFF	OFF	ON	OFF
X	4	6	OFF	ON	ON	OFF	OFF	OFF	ON	OFF
X	4	7	ON	ON	ON	OFF	OFF	OFF	ON	OFF
X	4	8	OFF	OFF	OFF	ON	OFF	OFF	ON	OFF
X	4	9	ON	OFF	OFF	ON	OFF	OFF	ON	OFF
X	4	A	OFF	ON	OFF	ON	OFF	OFF	ON	OFF
X	4	B	ON	ON	OFF	ON	OFF	OFF	ON	OFF
X	4	C	OFF	OFF	ON	ON	OFF	OFF	ON	OFF
X	4	D	ON	OFF	ON	ON	OFF	OFF	ON	OFF
X	4	E	OFF	ON	ON	ON	OFF	OFF	ON	OFF
X	4	F	ON	ON	ON	ON	OFF	OFF	ON	OFF
X	5	0	OFF	OFF	OFF	OFF	ON	OFF	ON	OFF
X	5	1	ON	OFF	OFF	OFF	ON	OFF	ON	OFF
X	5	2	OFF	ON	OFF	OFF	ON	OFF	ON	OFF
X	5	3	ON	ON	OFF	OFF	ON	OFF	ON	OFF
X	5	4	OFF	OFF	ON	OFF	ON	OFF	ON	OFF
X	5	5	ON	OFF	ON	OFF	ON	OFF	ON	OFF
X	5	6	OFF	ON	ON	OFF	ON	OFF	ON	OFF
X	5	7	ON	ON	ON	OFF	ON	OFF	ON	OFF
X	5	8	OFF	OFF	OFF	ON	ON	OFF	ON	OFF
X	5	9	ON	OFF	OFF	ON	ON	OFF	ON	OFF
X	5	A	OFF	ON	OFF	ON	ON	OFF	ON	OFF
X	5	B	ON	ON	OFF	ON	ON	OFF	ON	OFF
X	5	C	OFF	OFF	ON	ON	ON	OFF	ON	OFF
X	5	D	ON	OFF	ON	ON	ON	OFF	ON	OFF
X	5	E	OFF	ON	ON	ON	ON	OFF	ON	OFF
X	5	F	ON	ON	ON	ON	ON	OFF	ON	OFF
X	6	0	OFF	OFF	OFF	OFF	OFF	ON	ON	OFF
X	6	1	ON	OFF	OFF	OFF	OFF	ON	ON	OFF
X	6	2	OFF	ON	OFF	OFF	OFF	ON	ON	OFF
X	6	3	ON	ON	OFF	OFF	OFF	ON	ON	OFF
X	6	4	OFF	OFF	ON	OFF	OFF	ON	ON	OFF
X	6	5	ON	OFF	ON	OFF	OFF	ON	ON	OFF
X	6	6	OFF	ON	ON	OFF	OFF	ON	ON	OFF
X	6	7	ON	ON	ON	OFF	OFF	ON	ON	OFF
X	6	8	OFF	OFF	OFF	ON	OFF	ON	ON	OFF
X	6	9	ON	OFF	OFF	ON	OFF	ON	ON	OFF
X	6	A	OFF	ON	OFF	ON	OFF	ON	ON	OFF
X	6	B	ON	ON	OFF	ON	OFF	ON	ON	OFF
X	6	C	OFF	OFF	ON	ON	OFF	ON	ON	OFF
X	6	D	ON	OFF	ON	ON	OFF	ON	ON	OFF
X	6	E	OFF	ON	ON	ON	OFF	ON	ON	OFF
X	6	F	ON	ON	ON	ON	OFF	ON	ON	OFF
X	7	0	OFF	OFF	OFF	OFF	ON	ON	ON	OFF
X	7	1	ON	OFF	OFF	OFF	ON	ON	ON	OFF
X	7	2	OFF	ON	OFF	OFF	ON	ON	ON	OFF
X	7	3	ON	ON	OFF	OFF	ON	ON	ON	OFF
X	7	4	OFF	OFF	ON	OFF	ON	ON	ON	OFF
X	7	5	ON	OFF	ON	OFF	ON	ON	ON	OFF
X	7	6	OFF	ON	ON	OFF	ON	ON	ON	OFF
X	7	7	ON	ON	ON	OFF	ON	ON	ON	OFF
X	7	8	OFF	OFF	OFF	ON	ON	ON	ON	OFF

Digit Number			STATUS OF RELAY							
Designates Module	Designates Status of Selected Module		WITH CODE AT LEFT DISPLAYED ON ANNUNCIATOR MODULE							
5-8 = RC8	2	3	1	2	3	4	5	6	7	8
X	7	9	ON	OFF	OFF	ON	ON	ON	ON	OFF
X	7	A	OFF	ON	OFF	ON	ON	ON	ON	OFF
X	7	B	ON	ON	OFF	ON	ON	ON	ON	OFF
X	7	C	OFF	OFF	ON	ON	ON	ON	ON	OFF
X	7	D	ON	OFF	ON	ON	ON	ON	ON	OFF
X	7	E	OFF	ON	ON	ON	ON	ON	ON	OFF
X	7	F	ON	ON	ON	ON	ON	ON	ON	OFF
X	8	0	OFF	OFF	OFF	OFF	OFF	OFF	OFF	ON
X	8	1	ON	OFF	OFF	OFF	OFF	OFF	OFF	ON
X	8	2	OFF	ON	OFF	OFF	OFF	OFF	OFF	ON
X	8	3	ON	ON	OFF	OFF	OFF	OFF	OFF	ON
X	8	4	OFF	OFF	ON	OFF	OFF	OFF	OFF	ON
X	8	5	ON	OFF	ON	OFF	OFF	OFF	OFF	ON
X	8	6	OFF	ON	ON	OFF	OFF	OFF	OFF	ON
X	8	7	ON	ON	ON	OFF	OFF	OFF	OFF	ON
X	8	8	OFF	OFF	OFF	ON	OFF	OFF	OFF	ON
X	8	9	ON	OFF	OFF	ON	OFF	OFF	OFF	ON
X	8	A	OFF	ON	OFF	ON	OFF	OFF	OFF	ON
X	8	B	ON	ON	OFF	ON	OFF	OFF	OFF	ON
X	8	C	OFF	OFF	ON	ON	OFF	OFF	OFF	ON
X	8	D	ON	OFF	ON	ON	OFF	OFF	OFF	ON
X	8	E	OFF	ON	ON	ON	OFF	OFF	OFF	ON
X	8	F	ON	ON	ON	ON	OFF	OFF	OFF	ON
X	9	0	OFF	OFF	OFF	OFF	ON	OFF	OFF	ON
X	9	1	ON	OFF	OFF	OFF	ON	OFF	OFF	ON
X	9	2	OFF	ON	OFF	OFF	ON	OFF	OFF	ON
X	9	3	ON	ON	OFF	OFF	ON	OFF	OFF	ON
X	9	4	OFF	OFF	ON	OFF	ON	OFF	OFF	ON
X	9	5	ON	OFF	ON	OFF	ON	OFF	OFF	ON
X	9	6	OFF	ON	ON	OFF	ON	OFF	OFF	ON
X	9	7	ON	ON	ON	OFF	ON	OFF	OFF	ON
X	9	8	OFF	OFF	OFF	ON	ON	OFF	OFF	ON
X	9	9	ON	OFF	OFF	ON	ON	OFF	OFF	ON
X	9	A	OFF	ON	OFF	ON	ON	OFF	OFF	ON
X	9	B	ON	ON	OFF	ON	ON	OFF	OFF	ON
X	9	C	OFF	OFF	ON	ON	ON	OFF	OFF	ON
X	9	D	ON	OFF	ON	ON	ON	OFF	OFF	ON
X	9	E	OFF	ON	ON	ON	ON	OFF	OFF	ON
X	9	F	ON	ON	ON	ON	ON	OFF	OFF	ON
X	A	0	OFF	OFF	OFF	OFF	OFF	ON	OFF	ON
X	A	1	ON	OFF	OFF	OFF	OFF	ON	OFF	ON
X	A	2	OFF	ON	OFF	OFF	OFF	ON	OFF	ON
X	A	3	ON	ON	OFF	OFF	OFF	ON	OFF	ON
X	A	4	OFF	OFF	ON	OFF	OFF	ON	OFF	ON
X	A	5	ON	OFF	ON	OFF	OFF	ON	OFF	ON
X	A	6	OFF	ON	ON	OFF	OFF	ON	OFF	ON
X	A	7	ON	ON	ON	OFF	OFF	ON	OFF	ON
X	A	8	OFF	OFF	OFF	ON	OFF	ON	OFF	ON
X	A	9	ON	OFF	OFF	ON	OFF	ON	OFF	ON
X	A	A	OFF	ON	OFF	ON	OFF	ON	OFF	ON
X	A	B	ON	ON	OFF	ON	OFF	ON	OFF	ON
X	A	C	OFF	OFF	ON	ON	OFF	ON	OFF	ON
X	A	D	ON	OFF	ON	ON	OFF	ON	OFF	ON
X	A	E	OFF	ON	ON	ON	OFF	ON	OFF	ON
X	A	F	ON	ON	ON	ON	OFF	ON	OFF	ON
X	B	0	OFF	OFF	OFF	OFF	ON	ON	OFF	ON
X	B	1	ON	OFF	OFF	OFF	ON	ON	OFF	ON
X	B	2	OFF	ON	OFF	OFF	ON	ON	OFF	ON
X	B	3	ON	ON	OFF	OFF	ON	ON	OFF	ON
X	B	4	OFF	OFF	ON	OFF	ON	ON	OFF	ON
X	B	5	ON	OFF	ON	OFF	ON	ON	OFF	ON
X	B	6	OFF	ON	ON	OFF	ON	ON	OFF	ON
X	B	7	ON	ON	ON	OFF	ON	ON	OFF	ON
X	B	8	OFF	OFF	OFF	ON	ON	ON	OFF	ON

Digit Number			STATUS OF RELAY							
Designates Module	Designates Status of Selected Module		WITH CODE AT LEFT DISPLAYED ON ANNUNCIATOR MODULE							
5-8 = RC8	2	3	1	2	3	4	5	6	7	8
X	B	9	ON	OFF	OFF	ON	ON	ON	OFF	ON
X	B	A	OFF	ON	OFF	ON	ON	ON	OFF	ON
X	B	B	ON	ON	OFF	ON	ON	ON	OFF	ON
X	B	C	OFF	OFF	ON	ON	ON	ON	OFF	ON
X	B	D	ON	OFF	ON	ON	ON	ON	OFF	ON
X	B	E	OFF	ON	ON	ON	ON	ON	OFF	ON
X	B	F	ON	ON	ON	ON	ON	ON	OFF	ON
X	C	0	OFF	OFF	OFF	OFF	OFF	OFF	ON	ON
X	C	1	ON	OFF	OFF	OFF	OFF	OFF	ON	ON
X	C	2	OFF	ON	OFF	OFF	OFF	OFF	ON	ON
X	C	3	ON	ON	OFF	OFF	OFF	OFF	ON	ON
X	C	4	OFF	OFF	ON	OFF	OFF	OFF	ON	ON
X	C	5	ON	OFF	ON	OFF	OFF	OFF	ON	ON
X	C	6	OFF	ON	ON	OFF	OFF	OFF	ON	ON
X	C	7	ON	ON	ON	OFF	OFF	OFF	ON	ON
X	C	8	OFF	OFF	OFF	ON	OFF	OFF	ON	ON
X	C	9	ON	OFF	OFF	ON	OFF	OFF	ON	ON
X	C	A	OFF	ON	OFF	ON	OFF	OFF	ON	ON
X	C	B	ON	ON	OFF	ON	OFF	OFF	ON	ON
X	C	C	OFF	OFF	ON	ON	OFF	OFF	ON	ON
X	C	D	ON	OFF	ON	ON	OFF	OFF	ON	ON
X	C	E	OFF	ON	ON	ON	OFF	OFF	ON	ON
X	C	F	ON	ON	ON	ON	OFF	OFF	ON	ON
X	D	0	OFF	OFF	OFF	OFF	ON	OFF	ON	ON
X	D	1	ON	OFF	OFF	OFF	ON	OFF	ON	ON
X	D	2	OFF	ON	OFF	OFF	ON	OFF	ON	ON
X	D	3	ON	ON	OFF	OFF	ON	OFF	ON	ON
X	D	4	OFF	OFF	ON	OFF	ON	OFF	ON	ON
X	D	5	ON	OFF	ON	OFF	ON	OFF	ON	ON
X	D	6	OFF	ON	ON	OFF	ON	OFF	ON	ON
X	D	7	ON	ON	ON	OFF	ON	OFF	ON	ON
X	D	8	OFF	OFF	OFF	ON	ON	OFF	ON	ON
X	D	9	ON	OFF	OFF	ON	ON	OFF	ON	ON
X	D	A	OFF	ON	OFF	ON	ON	OFF	ON	ON
X	D	B	ON	ON	OFF	ON	ON	OFF	ON	ON
X	D	C	OFF	OFF	ON	ON	ON	OFF	ON	ON
X	D	D	ON	OFF	ON	ON	ON	OFF	ON	ON
X	D	E	OFF	ON	ON	ON	ON	OFF	ON	ON
X	D	F	ON	ON	ON	ON	ON	OFF	ON	ON
X	E	0	OFF	OFF	OFF	OFF	OFF	ON	ON	ON
X	E	1	ON	OFF	OFF	OFF	OFF	ON	ON	ON
X	E	2	OFF	ON	OFF	OFF	OFF	ON	ON	ON
X	E	3	ON	ON	OFF	OFF	OFF	ON	ON	ON
X	E	4	OFF	OFF	ON	OFF	OFF	ON	ON	ON
X	E	5	ON	OFF	ON	OFF	OFF	ON	ON	ON
X	E	6	OFF	ON	ON	OFF	OFF	ON	ON	ON
X	E	7	ON	ON	ON	OFF	OFF	ON	ON	ON
X	E	8	OFF	OFF	OFF	ON	OFF	ON	ON	ON
X	E	9	ON	OFF	OFF	ON	OFF	ON	ON	ON
X	E	A	OFF	ON	OFF	ON	OFF	ON	ON	ON
X	E	B	ON	ON	OFF	ON	OFF	ON	ON	ON
X	E	C	OFF	OFF	ON	ON	OFF	ON	ON	ON
X	E	D	ON	OFF	ON	ON	OFF	ON	ON	ON
X	E	E	OFF	ON	ON	ON	OFF	ON	ON	ON
X	E	F	ON	ON	ON	ON	OFF	ON	ON	ON
X	F	0	OFF	OFF	OFF	OFF	ON	ON	ON	ON
X	F	1	ON	OFF	OFF	OFF	ON	ON	ON	ON
X	F	2	OFF	ON	OFF	OFF	ON	ON	ON	ON
X	F	3	ON	ON	OFF	OFF	ON	ON	ON	ON
X	F	4	OFF	OFF	ON	OFF	ON	ON	ON	ON
X	F	5	ON	OFF	ON	OFF	ON	ON	ON	ON
X	F	6	OFF	ON	ON	OFF	ON	ON	ON	ON
X	F	7	ON	ON	ON	OFF	ON	ON	ON	ON
X	F	8	OFF	OFF	OFF	ON	ON	ON	ON	ON

Digit Number			STATUS OF RELAY							
Designates Module	Designates Status of Selected Module		WITH CODE AT LEFT DISPLAYED ON ANNUNCIATOR MODULE							
5-8 = RC8	2	3	1	2	3	4	5	6	7	8
X	F	9	ON	OFF	OFF	ON	ON	ON	ON	ON
X	F	A	OFF	ON	OFF	ON	ON	ON	ON	ON
X	F	B	ON	ON	OFF	ON	ON	ON	ON	ON
X	F	C	OFF	OFF	ON	ON	ON	ON	ON	ON
X	F	D	ON	OFF	ON	ON	ON	ON	ON	ON
X	F	E	OFF	ON	ON	ON	ON	ON	ON	ON
X	F	F	ON	ON	ON	ON	ON	ON	ON	ON

Exhibit 5-15 RC8 Status Codes
Example Status Displays:

“500” – The first RC8, #5, has all relays OFF.

“7A2” – The third RC8, #7, has Relay 1 OFF; Relay2 ON; Relay 3 OFF; Relay 4 OFF; Relay 5 OFF; Relay 6 ON; Relay 7 OFF; Relay 8 ON

X A 2 OFF ON OFF OFF OFF ON OFF ON

“813” – The fourth RC8, #8, has Relay 1 ON; Relay 2 ON; Relay 3 OFF; Relay 4 OFF; Relay 5 ON; Relay 6 OFF; Relay 7 OFF; Relay 8 OFF

X 1 3 ON ON OFF OFF ON OFF OFF OFF

Note: Specific trouble types per module can be accessed via the Fike EP Works™ software.

5.4.4 History Operation

The History Menu is provided to annunciate the order of the event occurrence on the protection loop attached to the Annunciator Module. In order to view the current history of the protection loop, scroll from record to record using the left pushbutton. The history of the protection loop is displayed in order of occurrence using the status codes defined in section 5.4.3 (Digit 1 is used to identify the system module. Digits 2 and 3 are used to indicate the event of the selected module as identified in Exhibit 5-19).

It displays the last 40 status events with the most recent being at the top of the file, and the oldest event at the bottom. The top of the history file (most recent) is designated with three dashed lines at the top of the display as shown in exhibit 5-16. The bottom of the history file is designated with three dashed lines at the bottom of the display as shown in exhibit 5-17. Use the + key to scroll forward in the events. Use the - key to scroll backwards in events. Press both keys at the same time to move to the top of the file (see exhibit 5-18).

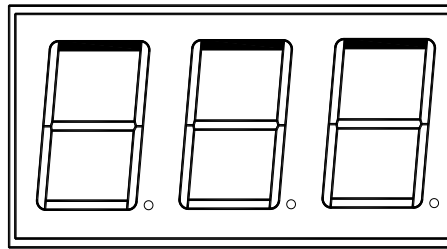


Exhibit 5-16 History Menu Top of File indicator

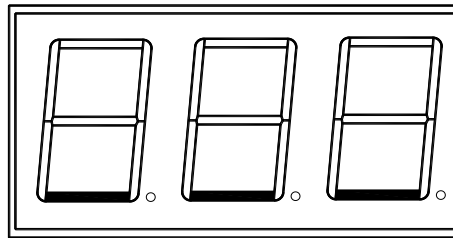


Exhibit 5-17 History Menu Bottom of File indicator

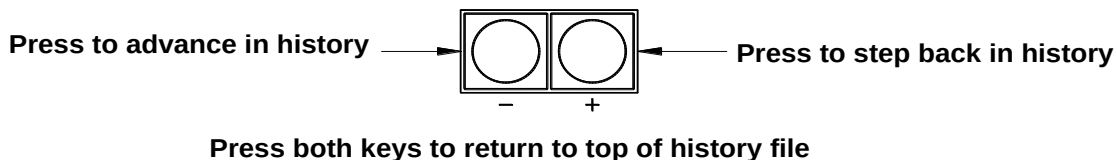


Exhibit 5-18 History Menu Navigation

Example History Messages:

"003" – An EPC on the Local Bus attached to the Annunciator Module has activated.

"102" – EPC #1 has a trouble on an input circuit.

Note: Specific trouble types per module can be accessed via the Fike EP Works software.

001	EPC #1 Not Connected	x02	EPC #x Trouble on Input Circuit
002	EPC #2 Not Connected	x03	EPC #x Pressure Warning Level Exceeded
003	EPC #3 Not Connected	x04	EPC #x Trouble on Output Circuit
004	EPC #4 Not Connected	x05	EPC #x Trouble on Fire Bus
005	RC8 #5 Not Connected	x06	EPC #x Predischage Condition Satisfied
006	RC8 #6 Not Connected	x07	EPC #x Supervisory Input Active
007	RC8 #7 Not Connected	x08	EPC #x Activated Releasing Outputs
008	RC8 #8 Not Connected	x09	EPC #x Disabled
009	PSU Not Connected	x10	EPC # x Failure during Arming
011	EPC #1 Connected	502	RC8 #5 Relay Status
012	EPC #2 Connected	602	RC8 #6 Relay Status
013	EPC #3 Connected	702	RC8 #7 Relay Status
014	EPC #4 Connected	802	RC8 #8 Relay Status
015	RC8 #1 Connected	901	PSU Normal Operation
016	RC8 #2 Connected	902	PSU Fan Trouble
017	RC8 #3 Connected	903	PSU Ground Fault
018	RC8 #4 Connected	904	PSU Battery or Charge Trouble
019	PSU Connected	905	PSU AC Trouble
01A	AM Power Up	906	PSU Battery Backup Being Utilized
01B	AM Power Off	907	PSU Output Trouble
01C	AM Serviced by Engineer	908	PSU Outputs Disabled
x01	EPC #x Normal Operation	909	PSU Auto Shutdown Mode

“x” = EPC 1 through 4

Exhibit 5-19 Display History Codes



Note: When a 903 PSU Ground Fault is received a Trouble will also be reported for one of the modules circuits connected on the Status Bus. This will provide indication as to the module and circuit that reported the ground fault condition.

5.4.5 Pressure Reading

Real time pressure can be accessed from any EPC connected to the AM via the Pressure Reading menu. This menu accesses the Detection 1 and Detection 2 circuits attached to each EPC. Note: This menu does not provide pressure reading access to any contact type detectors attached to the Detection 3 circuit.

Upon entry into the Pressure Menu (dEt), each detection circuit can be accessed by pressing the pushbutton. The left pushbutton scrolls backwards (8 to 1) and the right pushbutton scrolls forward (1 to 8). Display Digit 1 is used to designate the circuit from which the pressure is read. Digits 2 and 3 display the hexadecimal representation of the monitored pressure. Refer to Exhibit 5-21 for the correlation table.

Digit Number			
Designates EPC and Detection Input	Pressure (in hex form)*		
1	2	3	Detector Selected
1	X	X	EPC #1, Detection Circuit #1
2	X	X	EPC #1, Detection Circuit #2
3	X	X	EPC #2, Detection Circuit #1
4	X	X	EPC #2, Detection Circuit #2
5	X	X	EPC #3, Detection Circuit #1
6	X	X	EPC #3, Detection Circuit #2
7	X	X	EPC #4, Detection Circuit #1
8	X	X	EPC #4, Detection Circuit #2

Exhibit 5-20 Annunciator Module Pressure Menu Tables

Pressure Correlation Chart

Hexadecimal Display (Digits 2 & 3)	Pressure at Detector (PSI)	Pressure at Detector (mbar)	Hexadecimal Display (Digits 2 & 3)	Pressure at Detector (PSI)	Pressure at Detector (mbar)	Hexadecimal Display (Digits 2 & 3)	Pressure at Detector (PSI)	Pressure at Detector (mbar)
00	-5.23	-361	30	-2.80	-193	60	-1.07	-74
01	-5.18	-357	31	-2.77	-191	61	-1.03	-71
02	-5.07	-349	32	-2.74	-189	62	-1.00	-69
03	-4.95	-341	33	-2.69	-185	63	-0.96	-66
04	-4.85	-334	34	-2.65	-183	64	-0.92	-64
05	-4.77	-329	35	-2.62	-180	65	-0.89	-61
06	-4.70	-324	36	-2.58	-178	66	-0.86	-59
07	-4.63	-319	37	-2.54	-175	67	-0.82	-57
08	-4.56	-315	38	-2.52	-173	68	-0.79	-54
09	-4.50	-310	39	-2.48	-171	69	-0.75	-52
0A	-4.44	-306	3A	-2.43	-167	6A	-0.72	-49
0B	-4.39	-303	3B	-2.39	-165	6B	-0.68	-47
0C	-4.34	-299	3C	-2.36	-162	6C	-0.65	-45
0D	-4.28	-295	3D	-2.32	-160	6D	-0.61	-42
0E	-4.23	-292	3E	-2.28	-157	6E	-0.58	-40
0F	-4.18	-288	3F	-2.25	-155	6F	-0.55	-38
10	-4.13	-285	40	-2.21	-152	70	-0.51	-35
11	-4.08	-281	41	-2.17	-150	71	-0.47	-33
12	-4.04	-278	42	-2.13	-147	72	-0.44	-30
13	-4.00	-275	43	-2.10	-145	73	-0.40	-28
14	-3.94	-272	44	-2.06	-142	74	-0.37	-25
15	-3.90	-269	45	-2.03	-140	75	-0.34	-23
16	-3.86	-266	46	-1.99	-137	76	-0.31	-21
17	-3.81	-263	47	-1.96	-135	77	-0.27	-19
18	-3.77	-260	48	-1.92	-132	78	-0.23	-16
19	-3.72	-257	49	-1.88	-130	79	-0.20	-14
1A	-3.68	-254	4A	-1.85	-127	7A	-0.16	-11
1B	-3.64	-251	4B	-1.81	-125	7B	-0.13	-9
1C	-3.60	-248	4C	-1.77	-122	7C	-0.10	-7
1D	-3.56	-245	4D	-1.74	-120	7D	-0.06	-4
1E	-3.52	-242	4E	-1.71	-118	7E	-0.03	-2
1F	-3.47	-239	4F	-1.67	-115	7F	0.00	0
20	-3.43	-237	50	-1.63	-112	80	0.03	2
21	-3.39	-234	51	-1.59	-110	81	0.07	4
22	-3.35	-231	52	-1.56	-107	82	0.10	7
23	-3.31	-228	53	-1.52	-105	83	0.13	9
24	-3.26	-225	54	-1.49	-102	84	0.17	12
25	-3.22	-222	55	-1.45	-100	85	0.21	14
26	-3.18	-220	56	-1.42	-98	86	0.24	17
27	-3.14	-217	57	-1.39	-95	87	0.28	19
28	-3.10	-214	58	-1.35	-93	88	0.31	22
29	-3.06	-211	59	-1.31	-90	89	0.35	24
2A	-3.02	-208	5A	-1.28	-88	8A	0.38	26
2B	-2.99	-206	5B	-1.24	-86	8B	0.41	28
2C	-2.95	-203	5C	-1.21	-83	8C	0.45	31
2D	-2.91	-201	5D	-1.17	-81	8D	0.49	33
2E	-2.88	-198	5E	-1.14	-78	8E	0.52	36
2F	-2.84	-196	5F	-1.10	-76	8F	0.55	38

Exhibit 5-21 Pressure Correlation Chart

Pressure Tolerance: 0.9% of transducer span, 600 mbar (8.7 psi), using the CEREx™ pressure transducer, P/N 29945022-S, and EPC, P/N E10-0066, i.e. 5 mbar (0.08 psi).

Pressure Correlation Chart

Hexadecimal Display (Digits 2 & 3)	Pressure at Detector (PSI)	Pressure at Detector (mbar)
90	0.58	40
91	0.62	43
92	0.65	45
93	0.69	47
94	0.72	50
95	0.75	52
96	0.78	54
97	0.82	56
98	0.86	59
99	0.89	61
9A	0.92	63
9B	0.95	66
9C	0.99	68
9D	1.03	71
9E	1.06	73
9F	1.09	75
A0	1.13	78
A1	1.16	80
A2	1.19	82
A3	1.22	84
A4	1.26	87
A5	1.29	89
A6	1.33	91
A7	1.36	94
A8	1.40	96
A9	1.43	99
AA	1.46	101
AB	1.49	103
AC	1.53	105
AD	1.56	108
AE	1.59	110
AF	1.63	112
B0	1.66	115
B1	1.70	117
B2	1.73	119
B3	1.76	121
B4	1.80	124
B5	1.83	126
B6	1.86	128
B7	1.89	131
B8	1.93	133
B9	1.97	135
BA	2.00	138
BB	2.03	140
BC	2.06	142
BD	2.09	144
BE	2.13	147
BF	2.16	149

Hexadecimal Display (Digits 2 & 3)	Pressure at Detector (PSI)	Pressure at Detector (mbar)
C0	2.20	151
C1	2.23	154
C2	2.26	156
C3	2.29	158
C4	2.33	161
C5	2.36	163
C6	2.40	165
C7	2.43	167
C8	2.46	170
C9	2.50	172
CA	2.53	174
CB	2.56	177
CC	2.60	179
CD	2.63	181
CE	2.66	183
CF	2.69	186
D0	2.73	188
D1	2.76	190
D2	2.80	193
D3	2.83	195
D4	2.86	197
D5	2.90	200
D6	2.93	202
D7	2.96	204
D8	3.00	206
D9	3.03	209
DA	3.06	211
DB	3.09	213
DC	3.13	215
DD	3.16	218
DE	3.19	220
DF	3.23	222
E0	3.26	225
E1	3.29	227
E2	3.33	229
E3	3.36	232
E4	3.39	234
E5	3.42	236
E6	3.46	238
E7	3.49	241
E8	3.52	243
E9	3.56	245
EA	3.59	248
EB	3.62	250
EC	3.65	252
ED	3.69	254
EE	3.72	256
EF	3.75	259

Hexadecimal Display (Digits 2 & 3)	Pressure at Detector (PSI)	Pressure at Detector (mbar)
F0	3.79	261
F1	3.82	263
F2	3.85	266
F3	3.89	268
F4	3.92	270
F5	3.95	273
F6	3.98	275
F7	4.02	277
F8	4.05	279
F9	4.08	281
FA	4.12	284
FB	4.15	286
FC	4.18	288
FD	4.21	291
FE	4.25	293
FF	4.27	294

Exhibit 5-21 Pressure Correlation Chart (continued)

Pressure Tolerance: 0.9% of transducer span, 600 mbar (8.7 psi), using the CEREx™ pressure transducer, P/N 29945022-S, and EPC, P/N E10-0066, i.e. 5 mbar (0.08 psi).

5.4.6 Enable / Disable EPC

EPCs connected to the protection loop can be Enabled or Disabled directly from the AM via this menu option.



Note: Enabling or Disabling the EPC via this software method is only intended for use while the protected process is interrupted or idle.



WARNING: This mode of disabling is **NOT** intended for use while servicing the protected hazard! When servicing the protected hazard, power must be removed from the EPC, either directly at the power input to the EPC, or via the ON/OFF menu in section 5.4.7.

Upon entry into the Enable / Disable menu, each EPC can be accessed by pressing the left pushbutton. Display Digit 1 is used to designate the target EPC. Digit 2 displays the enabled or

disabled status of the selected EPC. Digit 3 displays the enabled or disabled command as being sent by the AM. Refer to Exhibit 5-23 for the ENABLED designation. Refer to Exhibit 5-24 for the DISABLED designation.

To change the status of the circuit, toggle digit #3 using the right button. It will change in status. After the status change has been verified, digit #2 will change to match digit #3. If it will not enable, check the EPC locally for proper power, configuration, and status bus communication.



WARNING: The EPC is not completely **ENABLED** or **DISABLED** until both digits #2 and #3 match.

Designates EPC	Designates Enabled/Disabled Status		EPC Selected to Enable/Disable
	1	2	
1	1	X	EPC #1
2	2	X	EPC #2
3	3	X	EPC #3
4	4	X	EPC #4

Exhibit 5-22 Enable/Disable Menu Table

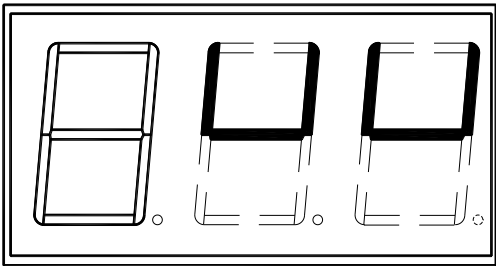


Exhibit 5-23 ENABLED Designation for Digits 2 and 3

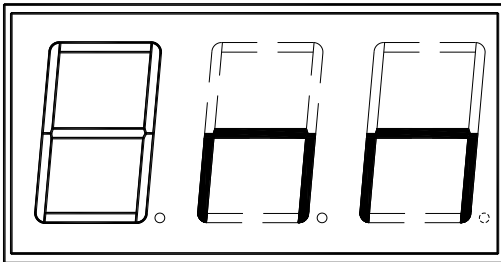


Exhibit 5-24 DISABLED Designation for Digits 2 and 3

5.4.7 PSU Output On/Off

PSU outputs 1 – 5 can be switched ON or OFF remotely from the Annunciator Module via this menu option.

 **Note:** This mode should be used to DISABLE EPCs when service is intended for the protected hazard!

Upon entry into the On / Off menu, each PSU power output circuit (1-5) can be accessed by pressing the left pushbutton. Display Digit 1 is used to designate the output circuit. Digit 2 displays the ON, or OFF status of the selected PSU output. Digit 3 displays the ON or OFF command as being sent by the AM. Refer to

Exhibit 5-26 for the ON designation. Refer to Exhibit 5-27 for the OFF designation.

To change the status of the circuit, toggle digit #3 using the right button. It will change in status. After the status change has been verified, digit #2 will change to match digit #3. If it will not enable, check the PSU locally for proper power, configuration, and status bus communication.

 **WARNING: The PSU output is not ON or OFF until both digits #2 and #3 match.**

PSU Output Circuit	On/Off Status of Output Circuit		PSU Output Selected
	2	3	
1	X	X	Output #1
2	X	X	Output #2
3	X	X	Output #3
4	X	X	Output #4
5	X	X	Output #5

Note: PSU Output 6 is not available to turn on or off.

Exhibit 5-25 PSU Power Output On/Off Table

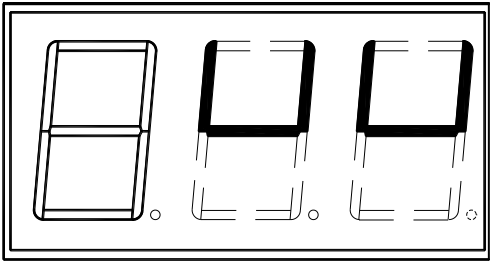


Exhibit 5-26 ON Designation for Digits 2 and 3

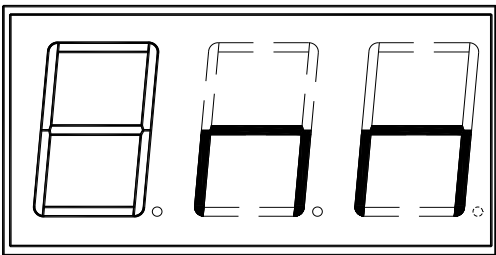


Exhibit 5-27 OFF Designation for Digits 2 and 3

5.4.8 Enable / Disable RC8

RC8's connected to the protection loop can be Enabled or Disabled directly from the AM via this menu option.

 **Note:** Enabling or Disabling the RC8 via this software method is only intended for use while the protected process is interrupted or idle.

 **WARNING:** This mode of disabling is **NOT** intended for use while servicing the protected hazard! When servicing the protected hazard, power must be removed from the RC8, either directly at the power input to the RC8, or via the ON/OFF menu in section 5.4.7.

Upon entry into the Enable / Disable menu, each RC8 can be accessed by pressing the left pushbutton. Display Digit 1 is used to designate

the target RC8. Digit 2 displays the enabled or disabled status of the selected RC8. Digit 3 displays the enabled or disabled command as being sent by the AM. Refer to Exhibit 5-29 for the ENABLED designation. Refer to Exhibit 5-30 for the DISABLED designation.

To change the status of the circuit, toggle digit #3 using the right button. It will change in status. After the status change has been verified, digit #2 will change to match digit #3. If it will not enable, check the RC8 locally for proper power, configuration, and status bus communication.

 **WARNING:** The RC8 is not completely **ENABLED** or **DISABLED** until both digits #2 and #3 match.

Designates RC8	Designates Enabled/Disabled Status		RC8 Selected to Enable/Disable
	1	2	
1	1	X	RC8 #1
2	2	X	RC8 #2
3	3	X	RC8 #3
4	4	X	RC8 #4

Exhibit 5-28 RC8 Enable/Disable Menu Table

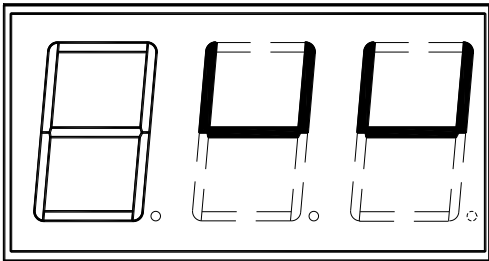


Exhibit 5-29 ENABLED Designation for Digits 2 and 3

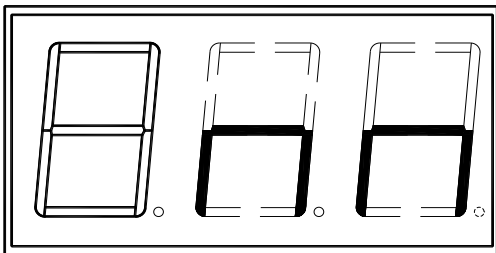


Exhibit 5-30 DISABLED Designation for Digits 2 and 3

5.4.9 Event History

The AM can store up to 40 history records in a First in First out (FIFO) table. A time stamp is created with each event recorded. The time information is updated each second and will be relative between consequent events.

The following codes are retrieved through EPWorks™ Software.

00-01	EPC #1 Not Connected	0x-02	EPC #x Trouble on Input Circuit
00-02	EPC #2 Not Connected	0x-03	EPC #x Pressure Warning Level Exceeded
00-03	EPC #3 Not Connected	0x-04	EPC #x Trouble on Output Circuit
00-04	EPC #4 Not Connected	0x-05	EPC #x Trouble on Fire Bus
00-05	RC8 #5 Not Connected	0x-06	EPC #x Predischage Condition Satisfied
00-06	RC8 #6 Not Connected	0x-07	EPC #x Supervisory Input Active
00-07	RC8 #7 Not Connected	0x-08	EPC #x Activated Releasing Outputs
00-08	RC8 #8 Not Connected	0x-09	EPC #x Disabled
00-09	PSU Not Connected	0x-10	EPC # x Failure during Arming
00-11	EPC #1 Connected	05-02	RC8 #5 Relay Status
00-12	EPC #2 Connected	06-02	RC8 #6 Relay Status
00-13	EPC #3 Connected	07-02	RC8 #7 Relay Status
00-14	EPC #4 Connected	08-02	RC8 #8 Relay Status
00-15	RC8 #1 Connected	09-01	PSU Normal Operation
00-16	RC8 #2 Connected	09-02	PSU Fan Trouble
00-17	RC8 #3 Connected	09-03	PSU Ground Fault
00-18	RC8 #4 Connected	09-04	PSU Battery or Charge Trouble
00-19	PSU Connected	09-05	PSU AC Trouble
00-1A	AM Power Up	09-06	PSU Battery Backup Being Utilized
00-1B	AM Power Off	09-07	PSU Output Trouble
00-1C	AM Serviced by Engineer	09-08	PSU Outputs Disabled
0x-01	EPC #x Normal Operation	09-09	PSU Auto Shutdown Mode

“x” = EPC 1 through 4

Exhibit 5-31 Event History Table

6.0 DECOMMISSIONING PROCEDURE / CHECK-LIST

The following procedure must only be performed by a Fike qualified Service Engineer, who has been assigned to prepare and complete the decommissioning of the above referenced Explosion Protection System.

Each step in the listed procedure must be adhered to and completion/acceptance of this form is mandatory. The Service Engineer must check off each of the following steps. In the case of non-compliance, the observed discrepancy must be corrected before completion of the system decommissioning.

STEPS / DESCRIPTION

1. Use Fike system/project component location diagram to record and verify the locations of all Fike system components for each zone and system.	☐ pass	☐ fail	☐ Remark/ Note No.
2. Control Panel to be disarmed/shutdown.	☐ pass	☐ fail	☐ Remark/ Note No.
3. All suppressor/valve actuators to be shunted.	☐ pass	☐ fail	☐ Remark/ Note No.
4. Each suppressor/valve container to be depressurised.  Warning: Do not ventilate nitrogen in a confined space.	☐ pass	☐ fail	☐ Remark/ Note No.
5. Power supply to the control panel to be Isolated by the customer and disconnected to prevent accidental reconnection. Fike to verify.	☐ pass	☐ fail	☐ Remark/ Note No.
6. Control panel battery to be disconnected and removed for disposal.	☐ pass	☐ fail	☐ Remark/ Note No.
7. Each actuator is to be removed and placed in a storage housing, to be either stored on site or removed for disposal.	☐ pass	☐ fail	☐ Remark/ Note No.
8. Verify that all suppressor/valve gauges are reading zero, replace fill valve cap loosely.	☐ pass	☐ fail	☐ Remark/ Note No.
ATTENTION: SYSTEM IS NOW DECOMMISSIONED AND READY FOR DISMOUNTING BY THE CUSTOMER.			

Exhibit 6-1 Decommissioning Steps

7.0 SPARE PARTS

Part Number	Description
E10-0068	Annunciator Module (AM)
02-4181	Fuse, 2A mini automotive blade type
02-10415	Six (6) Position Wire Terminal Block (P1 & P2)
02-10417	Three (3) Position Wire Terminal Block for Power Input (P3) & Trouble Relay (P4)
02-10876	End of Line Resistor, 140 Ω , 0.5 Watt for Bus (P1 & P2)

Exhibit 7-1 Spare Parts List

8.0 SPECIFICATIONS

Part Number:	E10-0068
Input Power:	18 to 30 VDC; 300mA
Max. Noise Ripple:	500mV peak at 24 VDC
Fuse:	2A Mini Automotive Type
Lower Limit for Input Power Shut-down:	16 VDC
Power Consumption:	130mA in Normal State
Terminal Block Capacity:	1.5 to 0.14 mm ² (16 to 28 AWG)
Relay Output:	One SPDT Form C for Trouble; One SPDT Form C for Alarm. Both rated DC operation: 2 amps @ 30VDC (pf=0.35) AC operation: 0.5 amps @ 250VAC (pf=0.35)
Status Bus:	Wire specification Alpha wire or RE-Y2Y or Belden 9841, maximum resistance of 50 Ω , inductance L=100uH, capacitance C=0.02uF, maximum length 300 m (1,000 ft) total.
Remote Bus:	Wire specification Alpha wire or RE-Y2Y or Belden 9841, maximum resistance of 50 Ω , inductance L=100uH, capacitance C=0.02uF, maximum length 300 m (1,000 ft) total.
Temperature Rating:	-18°C to 60°C (0°F to 140°F). If AM is not located inside auxiliary housing, maximum temperature is 43.3°C (110°F).
Humidity:	80% relative, maximum (non-condensing).
Size:	120mm L x 130mm H x 60mm D (4.8" L x 5.2" H x 2.3" D).
Weight:	0.5 kg (1 lbs.).
Hazardous Area:	The AM is not rated for hazardous atmospheres. The AM must be located in an auxiliary housing that is rated for the hazardous atmosphere in which it is located. To maintain the systems FM Approval for hazardous locations, the housing shall be an FM approved housing from the following: - E10-0085, E10-0087 or Killark XB-10126 or larger XB series housing approved for Class I, Div. 1 & 2, Groups C,D Class I, Zones 1 & 2, Groups IIB, IIA Class II, Div. 1 & 2, Groups E,F,G Class III NEMA 7(C,D), 9(E,F,G) - E10-0080, E10-0086, or Killark DB-10106 or larger DB series housing approved for Class II, Div. 1 & 2, Groups E,F,G Class III NEMA 3R, 4, 4X, 9(E,F,G) For ATEX installations the AM must be located in a non-hazardous atmosphere.
Current Firmware Version:	Can be verified using EPWorks™ Software.

9.0 REPAIR AND RETURN AUTHORIZATION

Any component that is to be returned to Fike must be approved for return prior to shipment. In order for the returned component to receive the correct attention, credit, repair, or replacement either under warranty or at the owner's expense, an authorization number must be assigned by Fike. A pre-arranged return authorization will expedite the business and corrective action measures taken upon receipt of the part(s).

A reference to the return authorization number should be inserted to the packing slip. If a packing slip is not used, then reference to the return authorization number should be made through alternate means. When preparing the component for shipment, please include your original Purchase Order Number, Invoice Number, or Fike Production Order Number. Include with the package the address you want the part shipped back to, shipment method, contact name, and telephone number.

A specific statement as the perceived defect or component failure will assist in examining the part(s). This statement should also address symptoms and an operation history of the system in which the component was installed.

In the event the suspect part is found within a larger top assembly component, the party assigning the Material Return Authorization Number (MRA) should be able to assist you as to whether the entire assembly must be returned or only the component in question.

To start the MRA procedure, contact the nearest Fike Sales Outlet. The nearest Fike Sales Outlet may be found on the Internet at www.fike.com. The nearest Fike Sales Outlet may also be found by calling one of the following:

AMERICAS

FIKE CORPORATION
USA
Phone +1-816-229-3405
Email: fpssales@fike.com

FIKE CANADA
Phone +1-905-681-3100
Email: fikecanada@fike.com

FIKE LATINA LTDA
Brazil
Phone +55-11-4525-5900
Email: fikelatina@fike.com

EUROPE

FIKE EUROPE
Belgium
Phone +32-14-21-00-31
Email: fike-europe@fike.com

FIKE UNITED KINGDOM
Phone +44-1-622-677081
Email: sales@fike.co.uk

FIKE FRANCE
Phone +33-1-30-31-31-32
Email: contact@fike.fr

FIKE DEUTSCHLAND
Germany
Phone +49-621-321-670
Email: info@fike.de

FIKE ITALIA
Phone +39-02-2952-4166
Email: webmaster@fike.it

FIKE IBERICA
Spain
Phone +34-93-6000-800
Email: fike@fike.es

FIKE IRELAND
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ASIA

FIKE ASIA PACIFIC SDN BHD
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FIKE JAPAN
Phone +81-335-951-291
Email: office@fikejapan.co.jp

FIKE INDIA
Phone +91-20-243-91-947
Email: fike.india@fike.com

MIDDLE EAST

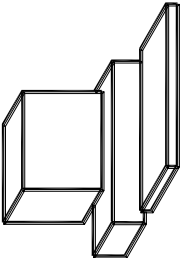
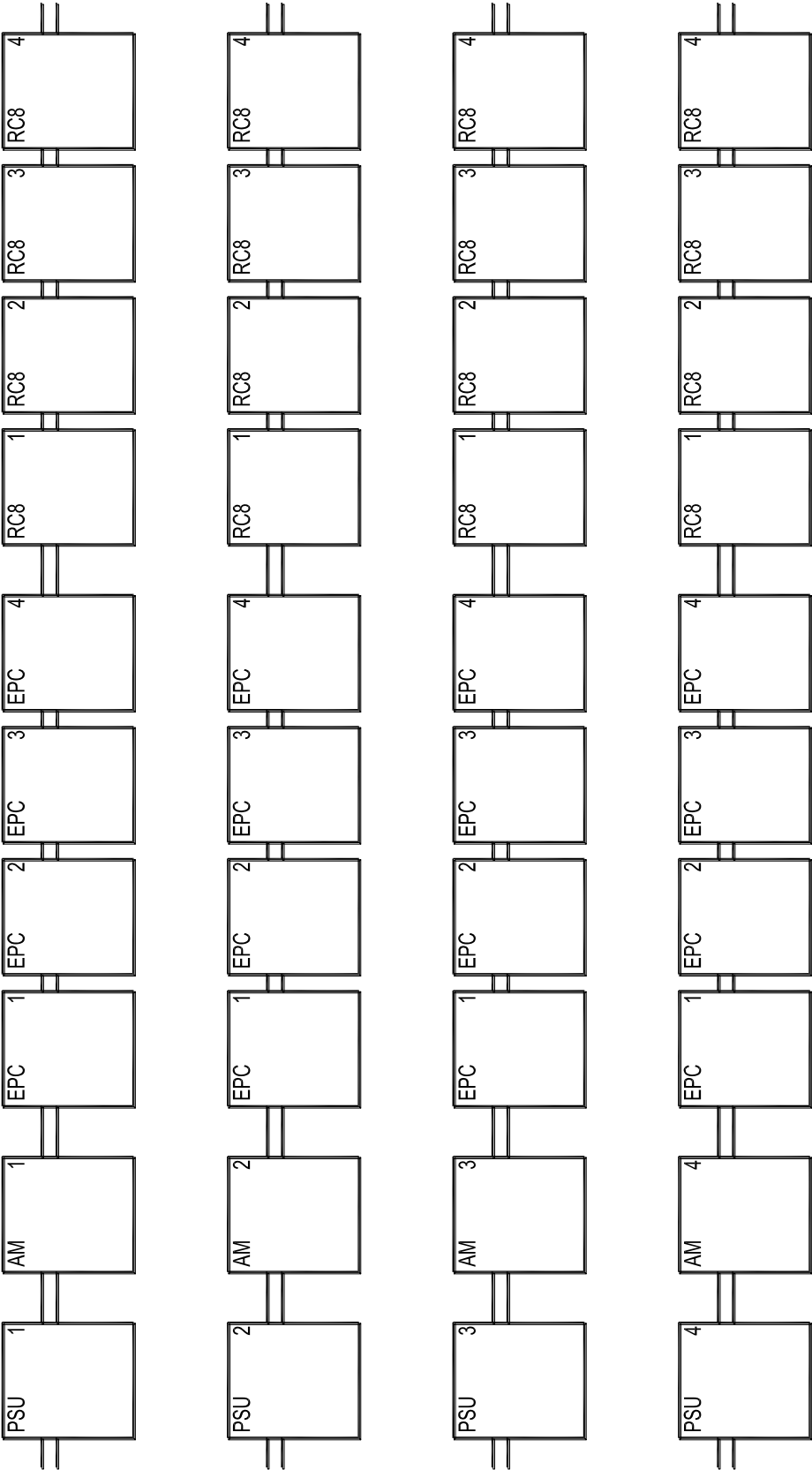
FIKE BAHRAIN
Dubai
Phone +971-40-386-6455
Email: fike.dubai@fike.com

ANNEX A EPACO WIRING DIAGRAM

This exhibit may be used to generate an overview of the EPACO system

(next page)

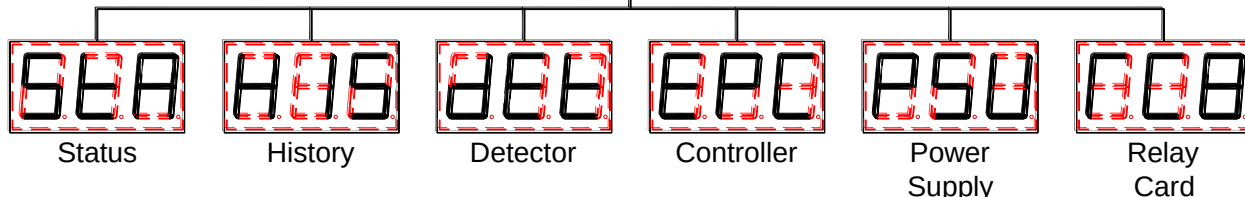
ANNEX A-EPACO WIRING DIAGRAM



Annex A EPACO Wiring Diagram

ANNEX B ANNUNCIATOR MODULE MENU
(next page)

Main Menu



Navigation

Display current Operating Mode:

Momentarily press both switches

Menu Mode:

Press and hold both switches for 5 seconds

Left Button (-): Scroll menu options

Right Button (+): Select desired option

Menu Mode Timeout: 20 seconds of inactivity



Status

Left Button (-): scrolls to next module

Right Button (+): scrolls to previous module

Annunciator Module

0	0	1	Normal Status Bus
0	0	2	Module Trouble on Status Bus
0	0	3	EP System on Status Bus Activated
0	0	4	Communication Failure on Status Bus
0	0	5	PC Mode

Explosion Protection Controller (EPC) X = EPC 1 to 4

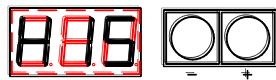
X	-	-	No EPC or EPC Shutdown
X	0	1	Normal Operation Mode
X	0	2	Trouble on one of the Input Circuits
X	0	3	Pressure Warning Level Exceeded
X	0	4	Trouble on one of the Output Circuits
X	0	5	Trouble on the Fire Bus
X	0	6	Predischage Condition Satisfied
X	0	7	Supervisory Input Active
X	0	8	EPC has Activated Releasing Outputs
X	0	9	EPC has been Disabled
X	1	0	Failure during Arming

Relay Card (RC8) X = RC8 5 to 8

X Y Y Y = Relay status Refer to Annunciator Manual E06-053 Section 5.4.3

Power Supply Unit (PSU)

9	-	-	No PSU or No Communications
9	0	1	Normal Operation
9	0	2	Fan Trouble
9	0	3	Ground Fault Trouble
9	0	4	Battery or Charge Trouble
9	0	5	AC Trouble
9	0	6	Battery Backup being Utilized
9	0	7	Output Trouble
9	0	8	Outputs Disabled
9	0	9	Shutdown Mode



History

Displays 40 most recent events

See Status Codes for definition of LEDs

Left Button (-): scrolls to older events

Right Button (+): scrolls to newer events

Pressing both buttons moves to the top of the file

Three lines across top of LEDs indicates Top of File (most recent)

Three lines across bottom of LEDs indicates Bottom of File



Pressure Detector

X X = Pressure Reading in Hex

Left Button (-): scrolls backward

Right Button (+): scrolls forward

1	X	X	EPC #1, Detection Circuit #1
2	X	X	EPC #1, Detection Circuit #2
3	X	X	EPC #2, Detection Circuit #1
4	X	X	EPC #2, Detection Circuit #2
5	X	X	EPC #3, Detection Circuit #1
6	X	X	EPC #3, Detection Circuit #2
7	X	X	EPC #4, Detection Circuit #1
8	X	X	EPC #4, Detection Circuit #2
6	2		-1.00 psig (-69 mbarg)
7	F		0.00 psig (0 mbarg)
8	D		0.49 psig (33 mbarg)
9	4		0.72 psig (50 mbarg)
9	C		0.99 psig (68 mbarg)

Refer to Annunciator Manual E06-053 Section 5.4.5 for additional values



Enabled/On



Disabled/Off

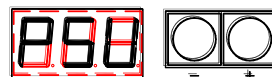


Explosion Protection Controller (EPC) X X = Enabled/Disabled

Left Button (-): scroll EPC selection

Right Button (+): toggle EPC state

1	X	X	EPC #1
2	X	X	EPC #2
3	X	X	EPC #3
4	X	X	EPC #4



Power Supply Unit (PSU) X X = On/Off

Left Button (-): scroll PSU device selection

Right Button (+): toggle PSU state

1	X	X	Output #1
2	X	X	Output #2
3	X	X	Output #3
4	X	X	Output #4
5	X	X	Output #5



Relay Card (RC8) X X = Enabled/Disabled

Left Button (-): scroll RC8 selection

Right Button (+): toggle RC8 state

5	X	X	RC8 #1
6	X	X	RC8 #2
7	X	X	RC8 #3
8	X	X	RC8 #4



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